10. (b) Weibull Distribution

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~~Aim~~ (keep MGF as future study)

(1. Core Concept)

The weibull distribution extends the exponential distribution to model scenarios where the event rate changes over time. It's widely used in reliability engineering and failure analysis.

Real world Analogy : High Bulb Lifespan Analysis

A manufacture produce a large batch of LED light bulbs. They want to estimate the failure pattern over time to improve quality control and offer an appropriate warranty.

* How Weibull Helps:

↳ Shape parameter $\beta < 1$ (e.g. 0.6).
→ Some bulbs fail very early due to manufacturing defects (infant mortality)
→ Suggest the need for better initial quality control.

↳ Shape parameter $\beta \approx 1$.
→ Bulbs fails at a constant rate over time, similar to Exponential distribution
→ Indicates random, memoryless failures not related to aging.

↳ Shape parameter $\beta > 1$ (e.g. 2.5).
→ Failure rate increases over time, similar to the Exponential distribution.
→ Suggest setting a warranty limit close to where most bulbs start failing.

Scale parameter η = 10,000 hours

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→ Failure rate increases over time, similar to the Exponential distribution.
→ Suggest setting a warranty limit close to when most bulbs start failing.

Scale parameter $\eta = 10,000$ hours
Means 63.2% of bulbs are expected to fail by 10,000 hours of use.

Why 63.2% fail by the scale Parameter (η)

→ In a weibull distribution, the scale (η) is not just a random number - it has a very specific interpretation

| When time $t = \eta$, the cumulative probability of failure (CDF) is 63.2% - regardless of the shape parameter (β)

Let's say
→ You're analyzing LED bulbs.
→ You find the scale parameter $\eta = 10,000$ hours.

Then,
| By 10,000 hours, about 63.2% of all bulbs are exposed to have failed.

It's not the average life, but more like the "characteristic life" - the point in time when failure becomes much more common.

Mathematically
The weibull CDF is: $F(t) = 1 - e^{-(t/\eta)^{\beta}}$.

Now plug in $t = \eta$:

$$F(\eta) = 1 - e^{-(\eta/\eta)^{\beta}} = 1 - e^{-1} \approx$$

$$1 - 0.3679 = 0.6321,$$

So, 63.2% of items fail by time $t = \eta$, always.

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(2. Key Parameters)

Parameter	Symbol	Role	Effect
Shape (β)	β	Controls failure rate behavior	- β < 1: Decreasing failure rate (early failure) - β ≈ 1: Constant rate (= Exponential). - β > 1: Increasing failure rate. (wear-out).
Scale (η)	η	Characteristic life time	Time by which 63.2% of units fail.

(3. Probability Density function (PDF))

(1) Theoretical Explanation

→ The weibull PDF describe the likelihood of failure at an exact time t. Its formula.

$$f(t) = \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1} e^{(-t/\eta)^{\beta}} \text{ for } \eta \geq 0.$$

→ β (shape): Controls failure behavior.
- β < 1: High early failure (e.g. manufacture defects)
- β = 1: Constant failure rate (= Exponential distribution)
- β > 1: Increase failure rate (e.g. wear-and-tear)

→ Govern the distribution's shape and failure rate behaviour. Also called the weibull slope.

→ η (scale): Time by which 63.2% of unit fails.
- ↳ Analogous to the "characteristic life" (where 63.2% of units fail by $t = \eta$)
- ↳ stretches or compresses the distribution horizontally.

2) Mathematical Derivation.

→ The Weibull PDF is derived from its CDF $F(t) = 1 - e^{-(t/\eta)^{\beta}}$ by differentiation.

$$f(t) = \frac{d}{dt} F(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta}\right)^{\beta-1} e^{-(t/\eta)^{\beta}}$$

→ The term $\left(\frac{t}{\eta}\right)^{\beta-1}$ captures time-dependent failure trends.

→ $e^{-(t/\eta)^{\beta}}$ ensures the PDF integrates to 1

3. Practical Examples

Scenario: Bearing with $\beta = 2.0$, $\eta = 8000$ hours
Compute $f(5000)$:

$$f(5000) = \frac{2}{8000} \left(\frac{5000}{8000}\right)^{1} e^{-(5000/8000)^{2}} \approx 0.000078$$

Interpretation: The instantaneous failure rate at 5000 hours is 0.0078% per hour.

Exercise 1: For $\beta = 1.5$, $\eta = 6000$ hours.

Q: Find the time 't' where the PDF peaks.

The Peak of the Weibull PDF

(1) Mathematical Derivation.

~~Start with the Weibull PDF:~~

Defn: The peak of the Weibull PDF is the point where the PDF reaches its maximum value. This is also referred to as the mode of distribution.

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$$f(t) = \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1} e^{-(t/\eta)^{\beta}}$$

Step 1: Take the natural logarithm (simplifies differentiation)

$$\ln f(t) = \ln\left(\frac{\beta}{\eta}\right) + (\beta-1)\ln\left(\frac{t}{\eta}\right) - \left(\frac{t}{\eta}\right)^{\beta}$$

Step 2: Differentiate w.r.t. t and set to zero:

$$\frac{d}{dt}\ln f(t) = \frac{\beta-1}{t} - \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1} = 0$$

Step 3: Solve for t

$$\frac{\beta-1}{t} = \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1}$$

$$t^{\beta} = \eta^{\beta}\left(\frac{\beta-1}{\beta}\right)$$

$$t_{peak} = \eta\left(\frac{\beta-1}{\beta}\right)^{1/\beta}$$

Condition

(a) t_{peak} only valid for $\beta > 1$

(b) For $\beta \leq 1$, the PDF has no peak (monotonically decreasing).

Interpretation

- The peak shift right as η increases (longer characteristic life).
- The peak becomes sharper as β increases.

Soln

$$t = 6000\left(\frac{0.5}{1.5}\right)^{1/1.5} \approx 3144 \text{ hours}.$$

(4) key Takeaways.

1. Shape (β): Determines if failure are early ($\beta < 1$), random ($\beta = 1$) or wear-out ($\beta > 1$).

2) Scale (η) : sets the characteristic lifetimes.

3) PDF Peak : Occurs at $t = \eta\left(\frac{\beta-1}{\beta}\right)^{1/\beta}$ (for $\beta > 1$).

(4. Cumulative Distribution Function)

1. Theoretical Explanation.

-> The weibull CDF $F(t)$ gives the probability that a system will by time 't'. It's S-shaped curve shows how failure probability accumulates over times, controlled by :

- Shape parameter (β) : Determines failure rate behavior.
- Scale parameter (η) : sets the characteristic lifetime (63.2% failure probability).

2. Mathematical formulation.

$$F(t) = 1 - e^{-(t/\eta)^{\beta}}.$$

Key Components :

(1) $(t/\eta)^{\beta}$: Time scaled by characteristic life, raised to β power.

(2) Negative exponent : Ensure $F(t)$ approaches 1 as $t \to \infty$.

(3) $1 - \ldots$ Converts survival probability to failure probability.

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③ Practical Exercise.

Exercise 1:

For $\beta = 1.8$, $\eta = 8000$ hours:

① Calculate $F(5000)$.

② Find time t where $F(t) = 0.5$ (median life).

① soln

$$F(5000) = 1 - e^{-(5000/8000)^{1.8}} = 0.413 \ (41.3\%).$$

② soln

$$1 - e^{-(t/8000)^{1.8}} = 0.5.$$

$$t = 8000 - (-\ln(0.5))^{1/1.8} = 6652 \text{ hours}.$$

Exercise 2:

A bearing has $\beta = 2.5$ / $\eta = 12000$ hours. What's the probability it survives beyond 10,000 hr?

soln

$$P(T > 10000) = 1 - F(1000) = e^{-(1000/12000)^{2.5}}$$

$$= 0.542 \ (54.2\%).$$

④ Properties

① Range $F(0) = 0$, $F(\infty) \to 1$.

② Inflection Point: where failure rate changes most rapidly.

- Occurs at $t = \eta \left(\frac{\beta - 1}{\beta}\right)^{1/\beta}$ (for $\beta > 1$.)

Inflection Point:

→ What is mean: The inflection point is where the CDF's curve changes from "getting steeper" to "leveling off" (for $\beta > 1$).

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→ Practical meaning: The time when the system transitions from low failure risk to high failure risk.

Formula:

$$t_{inflection} = \eta \left(\frac{\beta - 1}{\beta}\right)^{1/\beta} \text{ (only for } \beta > 1\text{)}.$$

Example:

For a car engine with:

- $\beta = 2.0$ (wear-out failures)
- $\eta = 100,000$ miles (characteristic life).

$$t_{inflection} = 100,000 \left(\frac{2-1}{2}\right)^{1/2} = 100,000 \times \sqrt{0.5} \approx 70,711 \text{ miles}.$$

Interpretation:

- Before 70,711 miles: Failure risk increases slowly.
- After 70,711 miles: Failure risk accelerates rapidly.

Why This matters:

Helps schedule ~~preventive~~ just before the inflection point to avoid high-risk periods.

(3) Median life: $t_{0.5} = \eta (\ln 2)^{1/\beta}$

Median life

What it Means

→ The time by which 50% of systems are expected to fail.

→ Unlike η (63.3% failure point), this is true "middle" of the distributions.

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Formula:

$$t_{0.5} = \eta (\ln 2)^{1/\beta}$$

where $\ln 2 = 0.693$

Examples

Same car engine ($\beta = 2.0$, $\eta = 100{,}000$ miles):

$$t_{0.5} = 100{,}000 \times (0.693)^{1/2}$$

$$\approx 100{,}100 \times 0.832 \approx 83{,}200 \text{ miles.}$$

Interpretation:

50% of engine fail before 83,200 miles and 50% fail after.

Comparison with inflection Point:

- Inflection (70,711 miles): when failure start to spike.
- Median (83,200 miles): when half have failed.

Media life formula derivative

cdf,

$$F(t) = 1 - e^{-\left(\frac{t - loc}{\eta}\right)^{\beta}}$$

To find the median life, set $F(t) = 0.5$:

$$0.5 = 1 - e^{-\left(\frac{t - loc}{\eta}\right)^{\beta}}$$

1) subtract both sides: $e^{-\left(\frac{t - loc}{\eta}\right)^{\beta}} = 0.5$

2) Take natural logarithm: $-\left(\frac{t - loc}{scale}\right)^{\beta} = \ln(0.5)$

3) Simplify: $\left(\frac{t - loc}{\eta}\right)^{\beta} = -\ln(0.5)$

4) solve for t: $t = loc + scale(\eta)(-\ln(0.5))^{1/\beta}$

· Median life $= loc + scale \cdot (\ln(2))^{1/\beta}$

Because,

$$-\ln(0.5) = \ln(2)$$

Weibull PDF Overview:

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$$f(t) = \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1} e^{-(t/\eta)^{\beta}} \text{ for } \eta \geq 0.$$

~~where:~~

~~β ≥~~

1. Variable and Parameters:

Symbol	Name	Role	Interpretation
~~x~~ t	Random variable	input to the functions	Represents the observed value (e.g. time until failure, wind speed). Must be ≥ 0.
η	Scale parameter	Stretches the distribution	Sets the "characteristic life" (63.2% of units fail by $t = \eta$. units match η (e.g. hours, km)
β	Shape parameter	Shape the distribution	Determine failure rate behavior:

- $k < 1$: Decreasing failure rate (infant mortality).
- $k = 1$: Constant rate (exponential distribution)
- $k > 1$: increasing rate (wear-out).

2) Breaking down the PDF formula:

1. Normalization Term $\left(\frac{\beta}{\eta}\right)$

→ Ensure the total area $\left(\int_0^{\infty} f(\eta)\, d\eta = 1\right)$ under the curve equals 1.

→ Adjust for the units η (e.g. if η is in hours, then $\frac{\beta}{\eta}$ is in hours^{-1})

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2. Power term $\left(\frac{t}{\eta}\right)^{\beta-1}$

- Controls how density changes with t

↳ $\beta-1 > 0$ $(\beta > 1)$: Density increases then decreases (unimodal).

↳ $\beta-1 = 0$ $(\beta = 0)$: Density decays exponentially (no peak).

↳ $\beta-1 < 0$ $(\beta < 1)$: Density blows up at $t = 0$ (high early failures)

3. ~~Power Term~~ Exponential Decay Term $\left(e^{-(t/\eta)^{\beta}}\right)$

→ Drives the PDF to 0 as $t \to \infty$.

→ The term $\left(\frac{t}{\eta}\right)^{\beta}$ is dimensionless, ensuring the exponent is unit-free.

3) Why scale (η) and shape (β) parameter?

1) Scale Parameter (η) : The 'Time Horizon'/'Characteristic Life'

- What it does:

→ η determines the scale or range over which event occur. It adjust the time axis, stretching or compressing the distribution.

→ η stretches or compresses the distribution horizontally (like zooming in/out on a timeline).

→ Think of it as the "expected life span" of a product or system.

→ It answers: "Over what time/range do events typically occur?"

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- Key Insight:
  → η is the point where 63.2% of all failures have occurred, regardless of the shape parameter β.

- 63.2% Rule (Why 63.2%?)
  - For $t = \eta$

$$F(\eta) = 1 - e^{-(\eta/\eta)^k} = 1 - e^{-1} \approx 0.632.$$

Example: η β.

- Suppose $\eta = 10$ hours and $k = 1$
  ↳ 63.2% of items are expected to fail by 10 hours.
- Now, if we double η to 20 hours
  ↳ 63.2% of failures occur by 100 hours, but the peak of the PDF may occur earlier or later depending on β.

2. Shape Parameter β: The 'Failure Pattern' / 'Failure Behavior Controller'

- What it does:
  → β controls how the failure rate changes over time. It determines the curve's shape (steep, flat or skewed)
  → β determines how failures evolve over time (increasing, decreasing, or constant rate).
  → It answers: "Do failures becomes more frequent, less frequent, or stay constant over time?".

β-value	Failure Rate Behavior	Real-world Analogy	Hazard function $h(x)$.
$\beta < 1$	Decreasing over time	"infant mortality": Early defects dominates	High at start, drops as weak units fail.
$\beta = 1$	Constant (Exponential)	Random failures (e.g. lightning	$\frac{1}{\eta}$ (Flat line).

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$\beta > 1$	increase over time	"Aging/wear-out (e.g metal fatigue)	Rises with t
$\beta = 2$	Linear increase (Rayleigh)	Stress accumulates linearly (e.g mechanical wear)	$\frac{2t}{\eta^2}$

3) Clarifying the Hazard function.

→ The hazard rate $h(t) = \left(\frac{\beta}{\eta}\left(\frac{t}{\eta}\right)\right)^{\beta-1}$ directly shows how β drives failure behavior.

- $k = 1$ (Exponential):

  $h(t) = \frac{1}{\eta}$ (constant - no aging effect).

- $k = 2$ (Rayleigh).

  $h(t) = \frac{2t}{\eta^2}$ (linear increase - stress builds over time)

- $k = 0.5$ (Infant Mortality).

  $h(t) = \frac{0.5}{\eta}\left(\frac{t}{\eta}\right)^{-0.5}$ (high at start, decreases)

~~4)~~ Summarize 'β'

Shape ("how")

- $k < 1$: Early failures. → $k = 0.8$: High infant mortality
- $k > 1$: Aging → $k = 3$: wear-out of the car tyre.
- ~~$k = 0.8$~~

4) Weibull PDF Interval Probability.

(a) By Integrating Pdf

$$P(a \le x \le b) = \int_a^b f(x; \beta, \eta)\, dx$$

$$= \int_a^b \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1} e^{-(t/\eta)^{\beta}}\, dt$$

$$\boxed{P(a \le x \le b) = e^{-t_a} - e^{-t_b}}$$

where

$$t_a = \left(\frac{a}{\eta}\right)^{\beta}$$

$$t_b = \left(\frac{b}{\eta}\right)^{\beta}$$

(b) By using cdf.

$$F(t; \beta, \eta) = 1 - e^{-(t/\eta)^{\beta}}$$

Thus, the probability over the interval $[a, b]$ can be expressed as the difference between the CDF values at b and a.

$$P(a \le X \le b) = F(b; \beta, \eta) - F(a; \beta, \eta)$$

Expanding the CDF terms:

$$P(a \le X \le b) = \left[1 - e^{-(b/\eta)^{\beta}}\right] - \left[1 - e^{-(a/\eta)^{\beta}}\right]$$

$$= e^{-(a/\eta)^{\beta}} - e^{-(b/\eta)^{\beta}}$$

5) Mean and Variance of the Weibull Distribution.

1) Mean (μ)

Formula:

$$\mu = \eta \Gamma\left(1 + \frac{1}{k}\right) = \eta F\left(1 + \frac{1}{\beta}\right)$$

- Explanation

→ The mean represent the expected value or average life of a random variable following weibull distribution.

→ The η parameter scales the mean, while β parameters adjusts how the distribution is shaped around the mean.

→ The Gamma function $F(z)$ generalize the factorial function for continuous values, extending the concept to non-integers and real values.

Intuitive Understanding:

- when $\beta = 1$, the weibull distribution reduces to the exponential distribution, and the mean become η
- when $\beta > 1$, the mean increases as more weight is given to values greater than η.
- When $\beta < 1$, the mean decreases as the distribution skews left, indicating more frequent early failures.

2. Variance (σ^2).

Formula

$$\sigma^2 = \eta^2 \left[\Gamma\left(1 + \frac{2}{\beta}\right) - \left(\Gamma\left(1 + \frac{1}{\beta}\right)\right)^2 \right]$$

- Explanation
  - The Variance measures the spread of the data around the mean.
  - It depends on both the scale parameter η and the shape parameter β, indicating how the distribution is stretched or compressed.
  - The first Gamma term $\Gamma\left(1 + \frac{2}{\beta}\right)$ captures the second moment, while the square of the mean term is subtracted to isolate the variance

Intuitive Understanding

- If $\beta = 1$, the variance simplifies to the exponential distribution's variance η^2
- For $\beta > 1$, variance decreases as the data becomes more concentrated around the mean (wear-out phase).
- For $\beta < 1$, variance increases as early failure. creates a long tail (infant mortality phase).

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Example of Mean and Variance Calculation for Weibull distribution.

Scenario:

- Scale parameter η = 1000 hours.
- Shape parameter β = 1.5.

We will calculate the mean μ and variance σ^2.

1. Mean Calculation

Formula:

$$\mu = \eta \Gamma\left(1 + \frac{1}{\beta}\right)$$

$$= 1000 \times \Gamma\left(1 + \frac{1}{1.5}\right).$$

$$= 1000 \times \Gamma(1.6667).$$

Now, we need to find $\Gamma(1.6667)$.
Using the Gamma function calculator or table:

$$\Gamma(1.6667) \approx 0.9027$$

Calculating the Mean

$$\mu = 1000 \times 0.9027 = 902.7 \text{ hours}.$$

2. Variance Calculation.

$$\sigma^2 = \eta^2 \left[\Gamma\left(1 + \frac{2}{\beta}\right) - \left(\Gamma\left(1 + \frac{1}{\beta}\right)\right)^2\right]$$

$$= 1000^2 \left[\Gamma(2.3333) - (\Gamma(1.6667))^2\right]$$

using the Gamma function calculator:

$$\Gamma(2.3333) \approx 1.1330.$$

$$[\Gamma(1.6667)]^2 \approx (0.9027)^2 \approx 0.8149$$

$$\sigma^2 = 1000^2 [1.1330 - 0.81149]$$

$$= 1000000 \times (0.3181)$$

$$= 318100 \text{ hours}^2$$

Weibull Survival Functions :

The survival function (SF), also called the Reliability function gives the probability that a random variable X exceed a specific value t.

Mathematically, the survival function $S_X(t)$ is defined as.

$$S_X(t) = P(X > t) = 1 - F_X(t).$$

where $F_X(t)$ is the CDF of the Weibull distribution.

The CDF of the Weibull distribution is:

$$F_X(t) = 1 - e^{-(t/\eta)^\beta}$$

Therefore, the Survival function is.

$$S_X(t) = 1 - (1 - e^{-(t/\eta)^\beta})$$

$$S_X(t) = e^{-(t/\eta)^\beta}, \quad t \geq 0$$

1) Import Properties of the Weibull Scenrival function

- At $t = 0$

$$S_X(0) = e^{-(0/\eta)^\beta} = e^0 = 1.$$

- As $\eta \to \infty$

$$S_X(\infty) = e^{-\infty} = 0.$$

- Median survival Time

→ $t_{0.5}$.

→ The median survival time $x_{0.6}$ is obtained by setting $S_X(t_{0.5}) = 0.5$

$$0.5 = e^{-(t_{0.5}/\eta)^{\beta}}$$

$$\ln(0.5) = -(t_{0.5}/\eta)^{\beta}$$

$$t_{0.5} = \eta \cdot (\ln(2))^{1/\beta}$$

→ The median survival time $t_{0.5}$ is the time by which 50% of the population is expected to survive.

→ Mathematically, it is the value of t where the SF $S_X(t)$ equal 0.5.

Properties summary

- As $t \to 0$, $S_X(t) \to 1$ (maximum reliability)
- As $t \to \infty$, $S_X(t) \to 0$ (complete failure)

2) Effect of Parameters η and β

- Scale Parameter η:

→ longer η shifts the SF to the right, indicating a longer survival time

- ~~Scale~~ Shape Parameter β:
  - $\beta < 1$: Decreasing hazard rate (early-life failure, rapid decrease in survival)
  - $\beta = 1$: Constant hazard rate (memoryless, exponential distributions)
  - $\beta > 1$: Increasing hazard rate (aging, wear out failure)

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3) Relation Between Survival Function and Hazard Function.

→ The hazards functions $h_x(x)$ represents the instantaneous failure rate at time t, given survival. survival until time t.

$$h_x(t) = \frac{f_x(t)}{S_x(t)}$$

$$= \frac{\frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1} e^{-(t/\eta)^{\beta}}}{e^{-(t/\eta)^{\beta}}}$$

$$= \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1}$$

Thus, the hazard function for the weibull distribution is a power law function, dependent only on β and η.

4) Example:

Problems:

A component has a Weibull distribution with $\eta = 1000$ hours and $\beta = 1.5$. we want to find the probability that the component survives beyond 800 hours.

$$S_x(800) = e^{-(800/1000)^{1.5}}$$

$$= e^{-0.612}$$

$$= 0.599$$

→ There is 59.9% probability that the component survives beyond 800 hours.

Weibull Distribution ①

Moment-Generating Function (MGF) of Weibull Distribution.

→ The Moment-Generating Function (MGF) of a random variable T is defined as

$$M_T(q) = E[e^{qt}] = \int_0^\infty e^{qt} f_T(t)\, dt$$

where

$$f_T(t) = \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1} e^{-(t/\eta)^{\beta}}$$ is Weibull pdf.

Notation Breakdown:

a) $M_T(q)$: The MGF of the random variable T.

→ The subscript T indicates that the function is specific to the distribution of T.

→ The argument q is a real valued parameter that 'shifts' the distribution to generate moments.

b) $E[\cdot]$: Expected Value Operator.

→ $E[e^{qt}]$ represents the expected value (or mean) of the expression e^{qt}

→ It is a way to calculate the average value of e^{qt} under the probability distribution defined by T.

c) e^{qt} : Exponential Term

→ e^{qt} is the key term in the MGF definition.
→ It introduces a transformation by multiplying the random variable t by q, then exponentiating it.

d) ~~$F_T(\pi)$~~ $F_T(t)$: Probability Density function (PDF)

→ $F_T(t)$, is the PDF of the random variable T.
→ It provide the likelihood of T taking a specific value t.
→ For the Weibull distribution, the PDF is given by

$$f_T(t) = \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1} e^{-(t/\eta)^{\beta}}.$$

e) q: Parameter of MGF.

→ q is a real-valued parameter that is not a characteristics of the random variable but rather a "dummy variable" introduced to generate moments.
→ By differentiating the MGF with respect to q, we can obtain the moment of the distribution.

1) First Moment (Mean).

Step 1

Define MGF

$$M_T(q) = \int_0^\infty e^{qt} \cdot \frac{\beta}{\eta} \left(\frac{t}{\eta}\right)^{\beta-1} e^{-(t/\eta)^\beta} dt$$

Combine the exponentials.

$$M_T(q) = \frac{\beta}{\eta} \int_0^\infty \left(\frac{t}{\eta}\right)^{\beta-1} \cdot e^{qt-(t/\eta)^\beta} dt$$

Step 2

1st derivative of MGF, differentiate with respect to q

$$M'_T(q) = \int_0^\infty t\, e^{qt} f_T(t)\, dt$$

$$= \frac{\beta}{\eta} \int_0^\infty t \left(\frac{t}{\eta}\right)^{\beta-1} e^{qt-(t/\eta)^\beta} dt$$

Now, set $q=0$

$$E[X] = M'_T(0) = \int_0^\infty t\, f_T(t)\, dt$$

This integral is known and simplifies to:

$$E[X] = \eta\, \Gamma\left(1+\frac{1}{\eta}\right)$$

2. Second Moment

The second moment $\mathbb{E}[\alpha T^2]$ is obtained by differentiating the MGF twice and setting $q = 0$.

Step 1

Second Derivative of MGF

$$M_T''(q) = \int_0^\infty t^2 e^{qt} f_T(t)\, dt$$

$$= \frac{\beta}{\eta} \int_0^\infty t^2 \left(\frac{t}{\eta}\right)^{\beta-1} e^{qt - (t/\eta)^\beta}\, dt$$

Step 2

Set $q = 0$:

$$\mathbb{E}[T^2] = M_T''(0) = \int_0^\infty t^2 f_t(t)\, dt$$

Step 3

This integral simplifies to:

$$\mathbb{E}[\alpha T^2] = \eta^2 \Gamma\left(1 + \frac{2}{\beta}\right)$$

Step 4

$$Var(T) = \mathbb{E}[\alpha T^2] - (\mathbb{E}[T])^2$$

$$= \alpha \eta^2 \left[\Gamma\left(1 + \frac{2}{\eta}\right) - \left(\Gamma\left(1 + \frac{1}{\eta}\right)\right)^2\right]$$

Special Case

→ When $\beta = 1$, the Weibull distribution reduces to the Exponential distribution and its MGF is:

$$M_T(q) = \frac{\eta}{\eta - q}, \quad q < \eta$$

→ This is only case where the Weibull MGF has a simple closed form.

Example: Calculate the first and second Momements of a Weibull Distribution.

Problem:

A random variable T follows a Weibull distribution with:

- Shape parameter $\beta = 2$
- Scale parameter $\eta = 3$.

Calculate the mean $E[T]$ and second moment $E[T^2]$

1. Mean $E[T]$

The mean of a Weibull distribution is given by

$$E[T] = 3 \cdot \Gamma\left(1 + \tfrac{1}{2}\right)$$

$$= 3 \cdot \Gamma(1.5).$$

Recall the relation

$$\Gamma(1.5) = \frac{\sqrt{\pi}}{2} = 0.8862.$$

$E[X] = 3 \times 0.8862 = 2.6586.$

2. Second Moment $E[T^2]$

$$E[T^2] = 3^2 \cdot \Gamma\left(1 + \frac{2}{2}\right).$$

$$= 9 \cdot \Gamma(2)$$

$$= 9 \times 1 = 9.$$

Summary

First Moment (Mean) → Central location / Expected value.

Second Moment → Spread and Variance.

What is Closed form?

→ A closed form is a mathematical expression that is expressed as a finite combination of basic functions and constants, such as

↳ Arthmetic Operations (additions, subtraction, multiplication, division)

↳ Elementary functions (polynomials, exponentials, logarithms, trigonometric functions).

↳ Well-known special functions (Gamma function, Beta function, etc)

→ A closed-form solution is concise and does not involve infinite sums, integrals, or iterative procedures.

Shortlog

- Simplicity : Easier to work with and interpret.
- Computation : Allows direct calculation without numerical methods.
- Analysis : Enables further mathematical manipulation such as differentiation and integration.

Why MGF is Not in Closed form?

→ The Weibull PDF has an exponent of the form $e^{-(t/\eta)^\beta}$, which makes the MGF integration complex and not expressible in a simple closed form.

→ Unlike simpler distribution (e.g. Exponential, Normal), the Weibull MGF cannot be easily simplified to a closed expression.

Therefore, we typically rely on moment definitions or use the first and second moments directly to derive the mean and variance.

Why is MGF important for Weibull?

→ Although we dont have a closed-form MGF, we can still derive moment using definition of the moment directly.

→ The lack of a closed-form MGF makes Weibull distribution challenging for

certain derivations, but the moment approach provides sufficient information for statistical analysis.

Weibull Distribution (2)

Python code

(1) Write a Program that calculate the Weibull Probability Density Function (PDF), probability interval, and the peak of the PDF (Tpeak) using manual calculation and built-in function from libraries like scipy?

```
Import numpy as np.
Import matplotlib.pyplot as plt.
From scipy.stats import weibull_min.
From scipy.integrate import quad.

# Parameters
lambda_param = 3  # Scale parameter
  K = 2  # Shape parameter.
  X = np.linspace (0, 20, 1000) # Range of x values.

# Manual calculation of weibull PDF.
  def weibull_pdf_manual(x, k, lambda_param)
      return (k / lambda_param ) * ( x /
      lambda_param) ** (k-1) * np.exp
       (-(x / lambda_param ) ** k ).

# Peak of PDF (Tpeak) calculation manually
def peak_pdf (k, lambda_param):
      if k <= 1:
          return 0 # Tpeak is at 0 if k <= 1
      return lambda_param * ((k-1)/k)**
          * (1/k)
```

```
# Manual PDF calculation.
pdf_manual = weibull_pdf_manual
                (x, k, lambda_param).

# Built-in Weibull PDF using scipy.
pdf_builtin = weibull_min.pdf(x, k,
                scale = lambda_param).

# Peak of PDF.
t_peak = peak_pdf(k, lamda_param)

# Interval Probability Calculation:
# Define the interval.
a, b = 2, 6.

# Using CDF to calculate interval
  probability
# P(a ≤ x ≤ b) = cdf(b) - cdf(a)
prob_interval = weibull_min.cdf(b, k,
                scale = lambda_param) -
                weibull_min.cdf(a, k, scale
                = lambda_param).

# Manual Calculation using integration of
  PDF
manual_interval_prob, _ = quad(weibull_
                pdf_manual, a, b, args = (k, lambda
                _param)).
```

output

T_peak : 2.1213
Interval prob: 0.6229.
Interval probability : 0.6225.

Note: There is no built-in function for T_peak.

(2) Write a program explaining weibull cdf and inflection point using both manual and built-in functions.

```
Import numpy as np.
import matplotlib.pyplot as plt
from scipy.stats import weibull_min

# Parameters
lambda_param = 3 # scale parameter
k = 2    # shape parameter
x = np.linspace(0, 10, 1000).

# Manual Calculation of weibull CDF
def weibull_cdf_manual(x, k, lambda_
                       param).
    return 1 - np.exp(-(x/lambda_param)
                      **k).
# Inflection point.
    if k<=1:
        return 0 # No inflection point for
    return lambda_param * ((k-1)/(k+1))
                 ** (1/k)
```

```
# Manual CDF calculation
cdf_manual = weibull_cdf_manual(x, k,
                lambda_param)

# Built-in weibull CDF using scipy.
cdf_builtin = weibull_min.cdf(x, k,
                scale = lambda_param).

# Calculate Inflection Point.
x_inf = inflection_point(k, lambda_param).
```

output

Inflection point : 1.7321.

(2) write a program explaining weibull survival functions and median survival time using both manual and built in functions.

```
import numpy as np.
import matplotlib.pyplot as plt.
from scipy.stats import weibull_min.

# Parameters
lambda_param = 5  # Scale parameter
k = 1.5  # shape parameter
x = np.linespace(0, 15, 1000).

# Manual calculation of weibull survival
function.
```

```
def weibull_survival_manual(x, k, lambda_param):
    return np.exp(-(x/lambda_param)**k)

# Median Survival Time Calculations
def median_survival_time(k, lambda_param):
    return lambda_param * (np.log(2))**(1/k)

# Manual survival function calculation
sf_manual = weibull_survival_manual(x, k, lambda_param)

# Built in survival function using scipy
sf_builtin = weibull_min.sf(x, k, scale=lambda_param)

# Manual Median Survival Time
Manual_x_median = median_survival_time(k, lambda_param)

# Builtin Median Survival Time
Builtin_x_median = weibull_min.median(k, scale=lambda_param)
```

Output

Median Survival Time: 4.1552

(4) Write a program calculating weibull mean and variance.

```
import numpy as np.
import matplotlib.pyplot as plt.
from scipy.stats import weibull_min
from scipy.special import gamma.

# Manual weibull Mean function.
def weibull_mean (k, lambda_param):
    return lambda_param * gamma(1+1/k)

# Manual weibull variance function.
def weibull_variance (k, lambda_param):
    term1 = gamma (1+2 (k).
    term2 = (gamma (1+1/k)) ** 2

    return ((lambda_param ** 2) *
            (term1 - term2)

# Calculate mean and variance using
  manual method.
mean_manual = weibull_mean (k, lamda-
                            param).
variance_manual = weibull_variance
                  (k, lambda_param)
# " " " using built-in function
mean_builtin = weibull_min.mean(k,
               scale = lambda_param).
varian_builtin = weibull_min.var(k,
                 scale = lambda_param)
```